

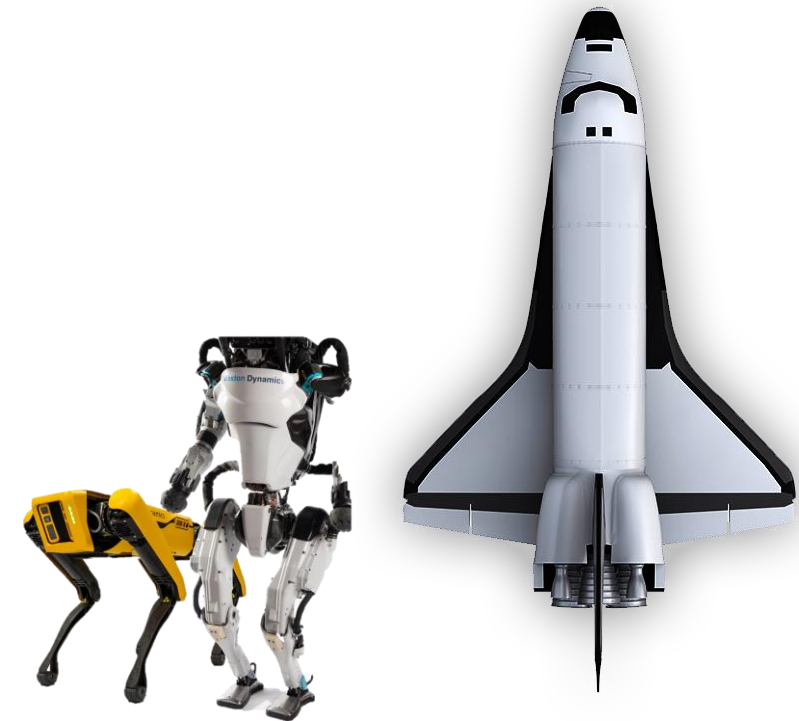
EEEE4064

ADVANCED CONTROL SYSTEM DESIGN

Lecture 2 State-Space Representation



University of
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Lecture Outline

01

State-Space Equations of LTI Systems

02

State-Space of a Second-Order System

03

State-Space of a Third-Order System

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State-Space of n^{th} -Order Systems – Case 1

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State-Space of n^{th} -Order Systems – Case 2

State-Space Equations of LTI Systems

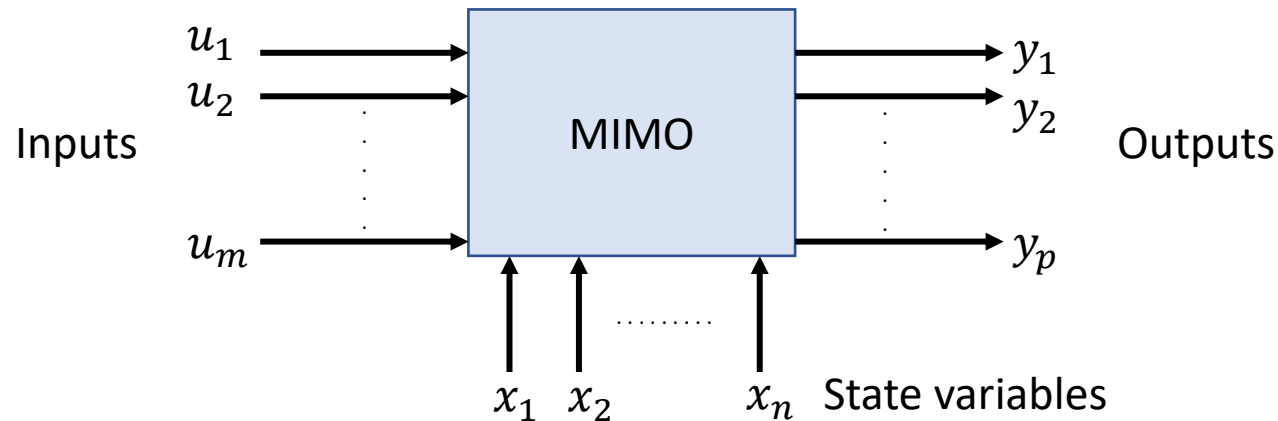
Previously, we have discussed the following:

01

The **state variables** of a dynamic system are the variables making up the smallest set of variables that determine the state of the dynamic system.

02

A **state-space representation** is a mathematical model of a physical system as a set of input, output and state variables related by first-order differential equations or difference equations.



So, how can we formulate the state-space representation? Let us see the following examples:

State-Space Equations of LTI Systems

EXAMPLE 1

Consider a continuous-time linear time-invariant system described by the following second-order differential equation:

$$\ddot{y}(t) + a_1 \dot{y}(t) + a_2 y(t) = u(t)$$

Find the state-space representation of the system when considering that a_1 and a_2 are constant system parameters,

$u(t)$ is the system input, $y(t)$ is the system output, $\dot{y}(t) = \frac{dy(t)}{dt}$, and $\ddot{y}(t) = \frac{d^2 y(t)}{dt^2}$.

Let:

$$x_1(t) = y(t)$$

$$x_2(t) = \dot{y}(t)$$

Differentiating $x_1(t)$ and $x_2(t)$ with respect to time t , respectively, we obtain:

State-Space Equations of LTI Systems

Differentiating $x_1(t)$ and $x_2(t)$ with respect to time, t , we obtain:

$$\dot{x}_1(t) = \dot{y}(t) = x_2(t) \quad (1)$$

$$\dot{x}_2(t) = \ddot{y}(t) = -a_2 x_1(t) - a_1 x_2(t) + u(t) \quad (2)$$

$$y(t) = x_1(t) \quad (3)$$

We can represent Equations (1) – (3) as follows:

$$\dot{x}_1(t) = \mathbf{0} \times x_1(t) + x_2(t) + \mathbf{0} \times u(t) \quad (4)$$

$$\dot{x}_2(t) = -a_2 x_1(t) - a_1 x_2(t) + u(t) \quad (5)$$

$$y(t) = x_1(t) + \mathbf{0} \times x_2(t) + \mathbf{0} \times u(t) \quad (6)$$

To convert to state space, Equations (4) – (6) are written in matrix form:

$$\begin{bmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -a_2 & -a_1 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u(t)$$

$$y(t) = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix}$$



$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}u$$

$$y = \mathbf{C}\mathbf{x} + Du$$

Output
equation

State
equation

- $x_1(t)$ and $x_2(t)$ are called the state variables.
- $\dot{x}_1(t)$, $\dot{x}_2(t)$, and $y(t)$ are called the state-space equations.
- The letters in **bold** font are **vectors** or **matrices**.

State-Space Equations of LTI Systems

Consider a **general** representation of a system using the following state-space representation:

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u}$$

$$\mathbf{y} = \mathbf{C}\mathbf{x} + \mathbf{D}\mathbf{u}$$

where:

$\mathbf{x}$ state vector

$\dot{\mathbf{x}}$ derivative of the state vector with respect to time

$\mathbf{y}$ output vector

$\mathbf{u}$ input or control vector

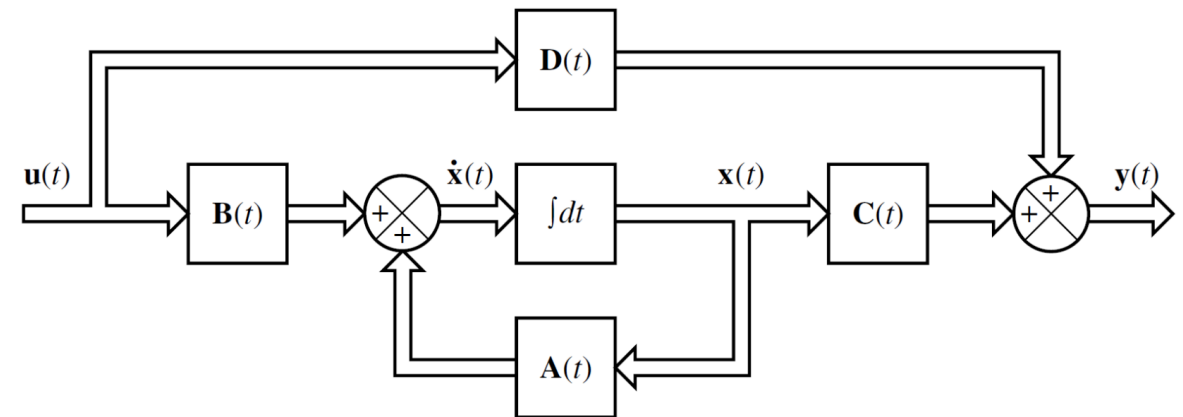
$\mathbf{A}$ system matrix

$\mathbf{B}$ input matrix

$\mathbf{C}$ output matrix

$\mathbf{D}$ feedforward matrix

- The system can be represented using the following vector block diagram:



Notes on the State-Space Representation

01

State variables do not need be physically measurable or observable quantities.

02

Practically, it is convenient to choose easily measurable quantities for the state variables, if this is possible, because optimal control laws will require the feedback of all state variables with suitable weighting.

03

For simplicity, it is possible to eliminate “ (t) ” when writing the state-space equations since it is known that we are dealing with time-domain.

04

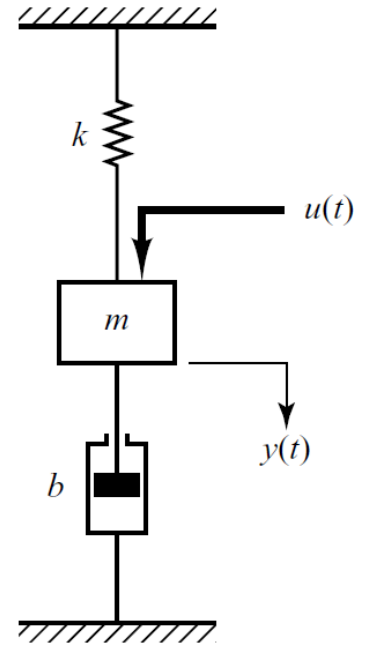
The state equation consists of n first-order differential equations, where n is equal to the order of the system.

State-Space of a Second-Order System

EXAMPLE 2

Consider the following mass-spring-damper system. It is assumed that the system is linear. The external force $u(t)$ is the input to the system, and the displacement $y(t)$ of the mass is the output. The displacement $y(t)$ is measured from the equilibrium position in the absence of the external force. This system is a single-input, single-output system.

Find the state-space representation of the system and sketch its vector block diagram.



01 Derive the second-order differential equation of the system.

02 Find 3 first-order differential equations based on the previous step, where 2 of them will be used for the state equation, and the last one will be used for the output equation.

State-Space of a Second-Order System

A mass-spring-damper can be represented by the following differential equation:

$$m\ddot{y} + b\dot{y} + ky = u$$

$$x_1 = y$$

$$x_2 = \dot{y}$$

$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = \ddot{y} = \frac{1}{m}(-ky - b\dot{y}) + \frac{1}{m}u$$

Rearranging:

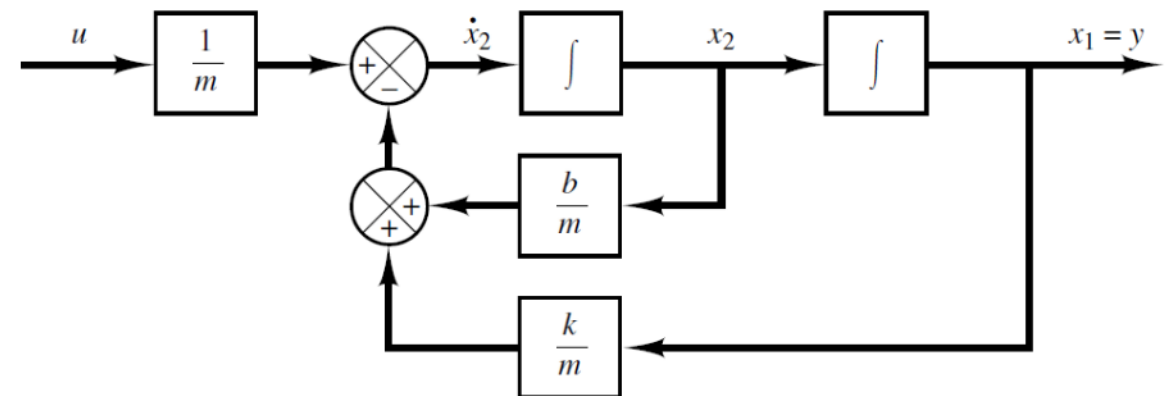
$$\dot{x}_2 = -\frac{k}{m}x_1 - \frac{b}{m}x_2 + \frac{1}{m}u$$

$$y = x_1$$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -\frac{k}{m} & -\frac{b}{m} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{1}{m} \end{bmatrix} u \iff \dot{\mathbf{x}} = \mathbf{Ax} + \mathbf{Bu}$$

$$y = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \iff y = \mathbf{Cx} + \mathbf{Du}$$

- The system can be represented using the following vector block diagram:



State-Space of a Third-Order System

EXAMPLE 3

Consider a system with the following differential equation:

$$\ddot{y}(t) + 6\dot{y}(t) + 11y(t) = 6u(t)$$

Find the state space representation for the system then draw its vector block diagram.

By rearranging the equation, we get:

$$\ddot{y} = -6\dot{y} - 11y + 6u$$

$$x_1 = \ddot{y}, \quad x_2 = \dot{y}, \quad x_3 = y$$

$$\dot{x}_1 = \ddot{y} = -6x_1 - 11x_2 - 6x_3 + 6u$$

$$\dot{x}_2 = \dot{y} = x_1$$

$$\dot{x}_3 = \dot{y} = x_2$$

$$y = x_3$$

Thus, the state-space representation is:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} -6 & -11 & -6 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 6 \\ 0 \\ 0 \end{bmatrix} u$$

$$y = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

State-Space of a Third-Order System

Thus:

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}u$$

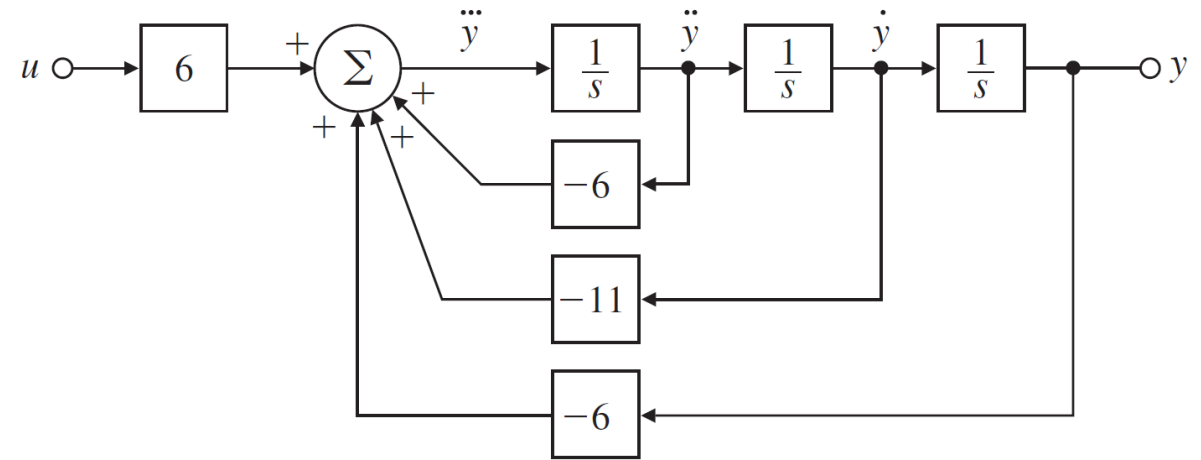
$$y = \mathbf{C}\mathbf{x} + Du$$

where

$$\mathbf{A} = \begin{bmatrix} -6 & -11 & -6 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 6 \\ 0 \\ 0 \end{bmatrix}$$

$$\mathbf{C} = [0 \quad 0 \quad 1], \quad D = 0$$

The vector block diagram of the system can be represented as follows:



State-Space of a Third-Order System

EXERCISE 1

Consider a system with the following differential equation:

$$\ddot{c} + 9\dot{c} + 26c = 24r$$

Find the state space representation for the system then draw its block diagram.

State-Space of n^{th} -Order Systems – Case 1

01

In **Case 1**, the forcing function does not involve derivative terms:

$$y^{(n)} + a_1 y^{(n-1)} + \cdots + a_{n-1} \dot{y} + a_n y = u$$

02

We may take the output and its derivations as a set of n state variables.

03

Mathematically, such a choice of state variables is quite convenient. Practically, however, because higher-order derivative terms are inaccurate, due to the noise effects inherent in any practical situations, such a choice of the state variables may not be desirable.

State-Space of n^{th} -Order Systems – Case 1

$$y^{(n)} + a_1 y^{(n-1)} + \dots + a_{n-1} \dot{y} + a_n y = u$$

Assume that:

$$x_1 = y$$

$$x_2 = \dot{y}$$

$\vdots$

$$x_n = y^{(n-1)}$$

Thus:

$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = x_3$$

$\vdots$

$$\dot{x}_{n-1} = x_n$$

$$\dot{x}_n = -a_n x_1 - \dots - a_1 x_n + u$$

The state-space representation is given as:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \vdots \\ \dot{x}_{n-1} \\ \dot{x}_n \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & \dots & 0 \\ 0 & 0 & 1 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 1 \\ -a_n & -a_{n-1} & -a_{n-2} & \dots & -a_1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_{n-1} \\ x_n \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix} u$$

$$y = \begin{bmatrix} 1 & 0 & \dots & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

$$\dot{\mathbf{x}} = \mathbf{Ax} + \mathbf{Bu}, \quad y = \mathbf{Cx}$$

The transfer function is given as:

$$\frac{Y(s)}{U(s)} = \frac{1}{s^n + a_1 s^{n-1} + \dots + a_{n-1} s + a_n}$$

State-Space of n^{th} -Order Systems – Case 2

01

In **Case 2**, the forcing function involves derivatives of the forcing function:

$$y^{(n)} + a_1 y^{(n-1)} + \cdots + a_{n-1} \dot{y} + a_n y = b_0 u^{(n)} + b_1 u^{(n-1)} + \cdots + b_{n-1} \dot{u} + b_n u$$

02

The main problem in defining the state variables for this case lies in the derivative terms of the input u .

03

The state variables must be such that they will eliminate the derivatives of u in the state equation.

State-Space of n^{th} -Order Systems – Case 2

$$y^{(n)} + a_1 y^{(n-1)} + \dots + a_{n-1} \dot{y} + a_n y = b_0 u^{(n)} + b_1 u^{(n-1)} + \dots + b_{n-1} \dot{u} + b_n u$$

To overcome the aforementioned issues, we define:

$$x_1 = y - \underline{\beta_0} u$$

$$x_2 = \dot{y} - \underline{\beta_0} \dot{u} - \underline{\beta_1} u = \dot{x}_1 - \beta_1 u$$

$$x_3 = \ddot{y} - \underline{\beta_0} \ddot{u} - \underline{\beta_1} \dot{u} - \underline{\beta_2} u = \dot{x}_2 - \beta_2 u$$

$\vdots$

$$x_n = y^{(n-1)} - \beta_0 u^{(n-1)} - \beta_1 u^{(n-2)} - \dots - \beta_{n-2} \dot{u} - \beta_{n-1} u = \dot{x}_{n-1} - \beta_{n-1} u$$

where

$$\underline{\beta_0} = b_0$$

$$\underline{\beta_1} = b_1 - a_1 \underline{\beta_0}$$

$$\underline{\beta_2} = b_2 - a_1 \underline{\beta_1} - a_2 \underline{\beta_0}$$

$$\beta_3 = b_3 - a_1 \underline{\beta_2} - a_2 \underline{\beta_1} - a_3 \underline{\beta_0}$$

$\vdots$

$$\beta_{n-1} = b_{n-1} - a_1 \beta_{n-2} - \dots - a_{n-2} \beta_1 - a_{n-1} \beta_0$$

Thus:

$$\dot{x}_1 = x_2 + \beta_1 u$$

$$\dot{x}_2 = x_3 + \beta_2 u$$

$\vdots$

$$\dot{x}_{n-1} = x_n + \beta_{n-1} u$$

$$\dot{x}_n = -a_n x_1 - a_{n-1} x_2 - \dots - a_1 x_n + \beta_n u$$

where

$$\beta_n = b_n - a_1 \beta_{n-1} - \dots - a_{n-1} \beta_1 - a_n \beta_0$$

- This choice of state variables provides a **uniqueness** of the solution of the state equation, but this is **not the only choice** of a set of state variables.
- With the present choice of state variables, we obtain the following state-space representation:

State-Space of n^{th} -Order Systems – Case 2

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \vdots \\ \dot{x}_{n-1} \\ \dot{x}_n \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \\ -a_n & -a_{n-1} & -a_{n-2} & \cdots & -a_1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_{n-1} \\ x_n \end{bmatrix} + \begin{bmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_{n-1} \\ \beta_n \end{bmatrix} u$$

$$y = \begin{bmatrix} 1 & 0 & \cdots & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} + \beta_0 u$$

$$\dot{\mathbf{x}} = \mathbf{Ax} + \mathbf{Bu}$$

$$y = \mathbf{Cx} + Du$$

The transfer function is given as:

$$\frac{Y(s)}{U(s)} = \frac{b_0 s^n + b_1 s^{n-1} + \cdots + b_{n-1} s + b_n}{s^n + a_1 s^{n-1} + \cdots + a_{n-1} s + a_n}$$

State-Space of n^{th} -Order Systems – Case 2

EXAMPLE 4

Find the state-space representation of the system defined by the following differential equation:

$$\ddot{y} + 2\dot{y} + 3y = 5\ddot{u} + 6\dot{u} + 7u + 8u$$

The general form that represents this system can be given as:

$$\ddot{y} + a_1\dot{y} + a_2y = b_0\ddot{u} + b_1\dot{u} + b_2u + b_3u$$

$$\beta_0 = b_0 = 5$$

$$\beta_1 = b_1 - a_1\beta_0 = 6 - 2 \times 5 = -4$$

$$\beta_2 = b_2 - a_1\beta_1 - a_2\beta_0 = 7 - 2(-4) - 3 \times 5 = 0$$

$$\beta_3 = b_3 - a_1\beta_2 - a_2\beta_1 - a_3\beta_0 = 8 - 2 \times 0 - 3(-4) - 4 \times 5 = 0$$

$$x_1 = y - \beta_0 u = y - 5u$$

$$x_2 = \dot{y} - \beta_0 \dot{u} - \beta_1 u = \dot{x}_1 - \beta_1 u = \dot{x}_1 + 4u$$

$$x_3 = \ddot{y} - \beta_0 \ddot{u} - \beta_1 \dot{u} - \beta_2 u = \dot{x}_2 - \beta_2 u = \dot{x}_2$$

$$\dot{x}_1 = x_2 + \beta_1 u = x_2 - 4u$$

$$\dot{x}_2 = x_3 + \beta_2 u = x_3$$

$$\dot{x}_3 = -a_3 x_1 - a_2 x_2 - a_1 x_3 + \beta_3 u = -4x_1 - 3x_2 - 2x_3$$

The state-space representation is given as follows:

State-Space of n^{th} -Order Systems – Case 2

$$\dot{\mathbf{x}} = \mathbf{Ax} + \mathbf{Bu}$$

$$y = \mathbf{Cx} + Du$$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -4 & -3 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} -4 \\ 0 \\ 0 \end{bmatrix} u$$

$$y = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + 5u$$

State-Space of n^{th} -Order Systems – Case 2

EXERCISE 2

Find the state-space equation and output equation for the system defined by:

$$\frac{Y(s)}{U(s)} = \frac{2s^3 + s^2 + s + 2}{s^3 + 4s^2 + 5s + 2}$$

Notes on the State-Space of n^{th} -Order Systems

01

Case 1: $y^{(n)} + a_1 y^{(n-1)} + \dots + a_{n-1} \dot{y} + a_n y = u$

02

Case 2: $y^{(n)} + a_1 y^{(n-1)} + \dots + a_{n-1} \dot{y} + a_n y = b_0 u^{(n)} + b_1 u^{(n-1)} + \dots + b_{n-1} \dot{u} + b_n u$

03

When comparing the state-space representation of Case 1 and Case 2, it can be seen that matrices **A** and **C** are exactly the same.

04

The derivatives of the input signal in Case 2 affect only the elements of the **B** matrix, while the gain of the highest derivative of the input affects matrix **D**.

Transformation of State-Space

Next Week ...

**DO YOU
HAVE ANY
QUESTIONS**